Lab 1 practice: separable and first-order linear DEs (Zill §2.2, §2.3)

Hand-solving drill for the Lab 1 quiz material. Closed book, about 10 minutes per problem. Problems 1 to 4 are separable, problems 5 to 8 are linear. Every solution was checked by substituting back into the ODE (`lab1_check.py`).

Separable. $\dfrac{dy}{dx} = g(x)h(y)$. Steps:

1. Find the constants $y = b$ where $h(b) = 0$. Each is a solution, set aside.
2. Otherwise divide: $\dfrac{dy}{h(y)} = g(x)\,dx$, integrate both sides, add one constant c.
3. Solve for y if possible. Check each constant solution against the general family: if no value of c produces it, it is a **lost (singular) solution** and must be listed separately.

Linear. Standard form (coefficient of y' equal to 1):

$$\frac{dy}{dx} + P(x)\,y = f(x)$$

Step	Formula
integrating factor	$\mu(x) = e^{\int P(x)\,dx}$ (drop the constant of integration)
multiply through	$\dfrac{d}{dx}[\mu y] = \mu f$
integrate	$\mu y = \displaystyle\int \mu f\,dx + c$, then divide by μ

Linear in x. If y appears nastily but x appears to the first power, flip it: $\dfrac{dx}{dy} + P(y)\,x = f(y)$.

Interval of validity. For a linear DE it is the largest interval containing the initial x on which P and f are continuous. For a separable DE, it is the largest interval containing the initial x on which the solution formula is defined and still satisfies the DE (watch square roots, logarithms, and denominators). Apply the initial condition to the **whole** general solution.

Problem 1

$$\frac{dy}{dx} = \frac{x^2}{y\,(1+x^3)}$$

Solution: separable, implicit answer

First order, and the right side is $g(x)\,h(y)$ with $g = \dfrac{x^2}{1+x^3}$ and $h = \dfrac{1}{y}$, so it is separable. There is no y with $h(y) = 0$, so no constant solutions to set aside.

Separate and integrate:

$$y\,dy = \frac{x^2}{1+x^3}\,dx \qquad \Longrightarrow \qquad \frac{y^2}{2} = \frac{1}{3}\ln|1+x^3| + c_1$$

since $\dfrac{d}{dx}(1+x^3) = 3x^2$. Multiply by 2 and absorb the constant:

$$\boxed{y^2 = \tfrac{2}{3}\ln|1+x^3| + c}$$

The solution is left implicit. It is valid wherever the right side is positive, and $y \neq 0$ is required because the ODE is undefined there.

Check: differentiate implicitly, $2yy' = \dfrac{2}{3}\cdot\dfrac{3x^2}{1+x^3}$, so $y' = \dfrac{x^2}{y(1+x^3)}$.

Problem 2

$\frac{dy}{dx} = x\,(y^2 - 4)$

Solution: separable with partial fractions and a lost solution

Right side is x times a function of y alone, so it is separable. $h(y) = y^2 - 4 = 0$ gives two constant solutions, $y = 2$ and $y = -2$. Set them aside.

Otherwise separate. Partial fractions: $\dfrac{1}{y^2-4} = \dfrac{1}{4}\left(\dfrac{1}{y-2} - \dfrac{1}{y+2}\right)$.

$$\frac{1}{4}\ln\left|\frac{y-2}{y+2}\right| = \frac{x^2}{2} + c_1 \qquad \Longrightarrow \qquad \frac{y-2}{y+2} = Ke^{2x^2}, \quad K = \pm e^{4c_1} \neq 0$$

Solve for y: $y - 2 = Ke^{2x^2}(y+2)$, so $y(1 - Ke^{2x^2}) = 2(1 + Ke^{2x^2})$.

Lost solution. Setting $K = 0$ in the family gives $y = 2$, so that constant is already included once K is allowed to be zero. But $y = -2$ would need $K \to \infty$. No finite K produces it, so it is lost and has to be stated separately.

$$\boxed{y = \frac{2(1 + Ke^{2x^2})}{1 - Ke^{2x^2}} \quad (K \text{ any constant}), \qquad \text{and also} \quad y = -2}$$

With $u = Ke^{2x^2}$ the family is $y = 2(1+u)/(1-u)$. It is defined for all x unless $u = 1$ somewhere, which happens only for $0 < K \leq 1$ (at $x = \pm\sqrt{\ln(1/K)/2}$). There the curve has a vertical asymptote and each branch is a separate solution on its own interval.

Check: $y = -2$ gives $y' = 0 = x(4-4)$. For the family, $u' = 4xu$, so $y' = \dfrac{4u'}{(1-u)^2} = \dfrac{16xu}{(1-u)^2}$, and $y^2 - 4 = \dfrac{16u}{(1-u)^2}$, so $y' = x(y^2-4)$.

Problem 3

$(1 + e^x)\, y \, \dfrac{dy}{dx} = e^x, \qquad y(0) = 1$

Solution: separable IVP with an interval of validity

Divide by $1+e^x$ (never zero): $\dfrac{dy}{dx} = \dfrac{e^x}{1+e^x}\cdot\dfrac{1}{y}$, which is separable. Separate and integrate:

$$y\,dy = \frac{e^x}{1+e^x}\,dx \qquad \Longrightarrow \qquad \frac{y^2}{2} = \ln(1+e^x) + c_1$$

The log needs no absolute value, since $1+e^x > 0$. Apply $y(0)=1$: $\frac{1}{2} = \ln 2 + c_1$, so $c_1 = \frac{1}{2} - \ln 2$. Then

$$y^2 = 1 + 2\ln\frac{1+e^x}{2}$$

The initial value $y(0)=1$ is positive, so take the positive root. Then for the solution to be real and nonzero (the ODE has y in the denominator after separating) we need

$$1 + 2\ln\frac{1+e^x}{2} > 0 \iff \frac{1+e^x}{2} > e^{-1/2} \iff x > \ln\!\left(2e^{-1/2}-1\right) \approx -1.546$$

$$\boxed{y = \sqrt{1 + 2\ln\frac{1+e^x}{2}}, \qquad x > \ln\!\left(2e^{-1/2}-1\right)}$$

At the left endpoint $y \to 0$, where $y' \to \infty$: the solution curve has a vertical tangent and cannot be continued, which is why the interval stops there.

Problem 4

$e^y \dfrac{dy}{dx} = x \sin x$

Solution: separable, integration by parts

Right side is a function of x alone times 1, left side is $e^y\,y'$, so it is separable. No constant solutions to set aside. Separate and integrate:

$$e^y\,dy = x\sin x\,dx$$

Left side integrates to e^y. For the right, integrate by parts with $u = x$, $dv = \sin x\,dx$, so $du = dx$ and $v = -\cos x$:

$$\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = \sin x - x\cos x$$

So $e^y = \sin x - x\cos x + c$. Take the log:

$$\boxed{y = \ln\left(\sin x - x\cos x + c\right)}$$

valid on any interval where $\sin x - x\cos x + c > 0$. The constant c decides where that is, so an initial condition $y(x_0) = y_0$ (which fixes $c = e^{y_0} - \sin x_0 + x_0\cos x_0$) also fixes the interval: the largest one around x_0 on which the argument of the log stays positive.

Check: $y' = \dfrac{x\sin x}{\sin x - x\cos x + c}$ (since $\dfrac{d}{dx}(\sin x - x\cos x) = x\sin x$), and $e^y y' = x\sin x$.

Problem 5

$x \dfrac{dy}{dx} + 4y = x^3 - x, \qquad x > 0$

Solution: linear, nonconstant coefficients

The unknown y and y' appear to the first power with coefficients depending on x only, so it is linear. It is not separable because of the $x^3 - x$ forcing term. Divide by x to reach standard form:

$$y' + \frac{4}{x}\, y = x^2 - 1$$

So $P = \dfrac{4}{x}$, $f = x^2 - 1$. Integrating factor for $x > 0$:

$$\mu = e^{\int 4/x\, dx} = e^{4 \ln x} = x^4$$

Multiply through; the left side collapses to a product derivative:

$$\frac{d}{dx}[x^4 y] = x^4(x^2 - 1) = x^6 - x^4 \qquad \Longrightarrow \qquad x^4 y = \frac{x^7}{7} - \frac{x^5}{5} + c$$

$$\boxed{y = \frac{x^3}{7} - \frac{x}{5} + c\, x^{-4}, \qquad 0 < x < \infty}$$

The interval is $(0, \infty)$ because $P = 4/x$ is not continuous at $x = 0$.

Check: $xy' = \dfrac{3x^3}{7} - \dfrac{x}{5} - 4cx^{-4}$ and $4y = \dfrac{4x^3}{7} - \dfrac{4x}{5} + 4cx^{-4}$. Adding gives $x^3 - x$.

Problem 6

$\frac{dy}{dx} + y = 5 \sin 2x$

Solution: linear, integration by parts twice

Already in standard form with $P = 1$, $f = 5\sin 2x$. Integrating factor $\mu = e^x$. Then

$$\frac{d}{dx}[e^x y] = 5e^x \sin 2x$$

Integrate $\int e^x \sin 2x\, dx$ by parts twice. Call it I.

$$I = e^x \sin 2x - 2\int e^x \cos 2x\, dx = e^x \sin 2x - 2\Big(e^x \cos 2x + 2\int e^x \sin 2x\, dx\Big)$$

$$I = e^x \sin 2x - 2e^x \cos 2x - 4I \qquad \Longrightarrow \qquad I = \frac{e^x(\sin 2x - 2\cos 2x)}{5}$$

So $e^x y = e^x(\sin 2x - 2\cos 2x) + c$. Divide by e^x:

$$\boxed{y = \sin 2x - 2\cos 2x + c\,e^{-x}, \qquad -\infty < x < \infty}$$

The ce^{-x} piece decays, so it is the transient. $\sin 2x - 2\cos 2x$ is the steady-state response.

Check: $y' = 2\cos 2x + 4\sin 2x - ce^{-x}$. Adding y gives $5\sin 2x$.

Problem 7

$\cos x \, \dfrac{dy}{dx} + (\sin x)\, y = 1, \qquad y(0) = 2$

Solution: linear IVP, trigonometric coefficients

Linear in y. Divide by $\cos x$ to reach standard form (valid where $\cos x \neq 0$):

$$y' + (\tan x)\, y = \sec x$$

$P = \tan x$ and $f = \sec x$. The integrating factor on $-\pi/2 < x < \pi/2$:

$$\mu = e^{\int \tan x\, dx} = e^{\ln \sec x} = \sec x$$

$$\frac{d}{dx}[y \sec x] = \sec^2 x \qquad \Longrightarrow \qquad y \sec x = \tan x + c$$

Multiply by $\cos x$: $y = \sin x + c \cos x$. Apply $y(0) = 2$: $0 + c = 2$, so $c = 2$.

$$\boxed{y = \sin x + 2\cos x, \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}}$$

The interval comes from the continuity of $P = \tan x$ and $f = \sec x$ around the starting point $x = 0$: they break at $\pm\pi/2$. The formula $\sin x + 2\cos x$ would itself make sense for all x, but the theorem only guarantees the DE in standard form holds on this interval.

Check: $y' = \cos x - 2\sin x$, so $\cos x\, y' + \sin x\, y = \cos^2 x - 2\sin x \cos x + \sin^2 x + 2\sin x \cos x = 1$.

Problem 8

$y\,dx - 4\,(x + y^6)\,dy = 0, \qquad y > 0$

Solution: linear in x, so write x as a function of y

In the form $\dfrac{dy}{dx} = \ldots$ this is nasty: $\dfrac{dy}{dx} = \dfrac{y}{4(x+y^6)}$ is neither separable nor linear in y. But x appears to the first power, so flip it. Divide by y and by dy:

$$\frac{dx}{dy} = \frac{4(x+y^6)}{y} \qquad \Longrightarrow \qquad \frac{dx}{dy} - \frac{4}{y}\,x = 4y^5$$

Now $x(y)$ is the unknown, $P = -\dfrac{4}{y}$, $f = 4y^5$. For $y > 0$:

$$\mu = e^{-4\int dy/y} = e^{-4\ln y} = y^{-4}$$

$$\frac{d}{dy}\big[y^{-4}x\big] = y^{-4}\cdot 4y^5 = 4y \qquad \Longrightarrow \qquad y^{-4}x = 2y^2 + c$$

$$\boxed{x = 2y^6 + c\,y^4, \qquad y > 0}$$

This is x as a function of y, which is an acceptable (explicit) solution of the original differential form.

Check: $\dfrac{dx}{dy} = 12y^5 + 4cy^3$ and $\dfrac{4(x+y^6)}{y} = \dfrac{4(3y^6 + cy^4)}{y} = 12y^5 + 4cy^3$.